

Bayes Statistics (Bayes Theorem)

Bayesian statistics is an approach to data analysis and parameter estimation based on **Bayes' theorem**.

Baye's Theorem

Probability $\begin{cases} \rightarrow \text{Independent Events} \\ \rightarrow \text{Dependent Events} \end{cases}$

① Independent Events

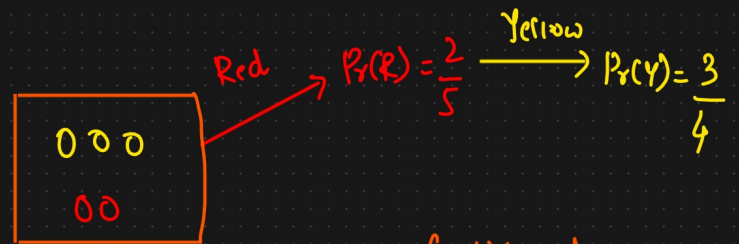
Eg: Rolling a dice
 $\{1, 2, 3, 4, 5, 6\}$

$$Pr(1) = \frac{1}{6} \quad Pr(2) = \frac{1}{6} \quad \dots$$

Tossing a coin

$$Pr(H) = 0.5 \quad Pr(T) = 0.5$$

② Dependent Event



$$\begin{aligned} Pr(R \text{ and } Y) &= P(R) * \boxed{Pr(Y/R)} \\ &= \frac{2}{5} * \frac{3}{4} = \frac{6}{20} // \end{aligned}$$

$\Rightarrow$ Conditional probability

$$Pr(A \text{ and } B) = Pr(B \text{ and } A)$$

$$Pr(A) * Pr(B/A) = Pr(B) * Pr(A/B)$$

$$\boxed{Pr(B/A) = \frac{Pr(B) * Pr(A/B)}{Pr(A)}} \Rightarrow \text{Bayes' theorem}$$



$$P_r(A/B) = \frac{P_r(A) * P_r(B/A)}{P_r(B)}$$

$A, B = \text{events}$

$P_r(A/B) = \text{Probability of A given B is true}$

$P_r(B/A) = \text{" B " A is true}$

$P_r(A), P_r(B) = \text{Independent probabilities of A and B}$

Example:

<u>DATASET</u>			
	↑↑ Independent		↑↑ o/p / dependent
Size of House	No. of Rooms	Location	Price
x_1	x_2	x_3	y

$$P_r(y/x_1, x_2, x_3) = \frac{P_r(y) * P_r(x_1, x_2, x_3/y)}{P_r(x_1, x_2, x_3)}$$



Bayes Theorem